

```

1  --s16_2
2
3  LIBRARY ieee;
4  USE ieee.std_logic_1164.ALL;
5  USE ieee.numeric_std.ALL;
6
7  ENTITY s16_3_1a IS
8      PORT (
9          clk : IN STD_LOGIC;
10         u : IN STD_LOGIC;
11         x : OUT STD_LOGIC_VECTOR(1 DOWNTO 0)
12     );
13 END ENTITY;
14
15 ARCHITECTURE behavioral OF s16_3_1a IS
16     SIGNAL q : STD_LOGIC_VECTOR(1 DOWNTO 0) := (OTHERS => '0');
17     SIGNAL a, b, c, d, e, f, g : STD_LOGIC;
18 BEGIN
19
20     --combinational logic
21     a <= u OR q(1);
22     b <= q(1) OR q(0);
23     c <= (NOT u) OR (NOT q(1)) OR (NOT q(0));
24     d <= a AND b AND c;
25
26     e <= (NOT u) AND q(0);
27     f <= u AND (NOT q(0));
28     g <= e OR f;
29
30     -- outputs
31     x <= q;
32
33     --flipflops
34     ff : PROCESS (clk)
35     BEGIN
36         IF rising_edge(clk) THEN
37             q(0) <= g;
38             q(1) <= d;
39         END IF;
40     END PROCESS;
41
42 END ARCHITECTURE behavioral;

```